TripCraft: A Benchmark for Spatio-Temporally Fine Grained Travel Planning

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Abstract

Existing travel planning benchmarks suffer from spatial inconsistencies, semi-synthetic data, and binary evaluation metrics that fail to capture the nuanced quality of real-world itineraries. We introduce *TripCraft*, a comprehensive benchmark for spatio-temporally fine-grained travel planning that addresses these limitations. *TripCraft*contains 1,000 travel queries across 140 U.S. cities, integrating real-world constraints including public transit schedules (GTFS), event calendars, and detailed attraction categorizations. Each query includes rich user personas and human-annotated gold itineraries with expert rationales. We propose five novel continuous evaluation metrics that assess temporal meal scheduling, attraction duration appropriateness, spatial transit feasibility, sequential ordering, and persona alignment. Our evaluation of state-of-the-art language models reveals that parameter-informed prompting substantially improves temporal scheduling quality, with 7-day trip meal scores improving from 0.61 to 0.80. However, this comes at the cost of reduced constraint satisfaction, highlighting important trade-offs between itinerary quality and feasibility. *TripCraft*establishes a new standard for evaluating AI travel planning systems with realistic, constraint-rich scenarios and nuanced quality metrics.

1 Introduction

Travel planning is a complex, multi-faceted task that requires reasoning about spatial, temporal, and personal constraints while optimizing for user preferences and practical feasibility. As large language models (LLMs) have shown remarkable capabilities in various planning tasks, there is growing interest in their application to automated travel itinerary generation. However, existing benchmarks for evaluating travel planning systems suffer from several critical limitations that hinder their ability to assess real-world applicability.

Current travel planning benchmarks, such as TravelPlanner (Xie et al., 2024) and TravelAgent (Chen et al., 2024), primarily rely on semi-synthetic data that exhibits spatial inconsistencies and lacks integration with real-world constraints such as public transit schedules, event calendars, and fine-grained attraction categorizations. Moreover, these benchmarks typically employ binary evaluation metrics that fail to capture the nuanced quality differences between travel itineraries. For instance, two plans might both satisfy basic constraints but differ significantly in their temporal appropriateness, spatial efficiency, or alignment with user preferences.

We address these limitations by introducing *TripCraft*, a comprehensive benchmark for spatio-temporally fine-grained travel planning that grounds itinerary evaluation in real-world, constraint-rich scenarios. *TripCraft*makes several key contributions:

Real-world data integration: Our benchmark incorporates authentic constraints from 140 U.S. cities, including GTFS public transit data, comprehensive event calendars, and detailed attraction categorizations spanning 15 types. This ensures that generated itineraries can be evaluated against practical feasibility requirements.

Rich user personas and expert annotations: Each of our 1,000 travel queries includes detailed user personas capturing traveler type, purpose, spending preferences, and location preferences. Human annotators with travel planning expertise created gold standard itineraries with detailed rationales, providing high-quality reference plans.

Continuous evaluation metrics: We introduce five novel metrics that provide continuous, rather than binary, assessments of itinerary quality: Temporal Meal Score (meal timing appropriateness), Temporal Attraction Score (activity duration vs. pace preference), Spatial Score (transit accessibility), Ordering Score (sequence efficiency), and Per-

sona Score (preference alignment).

Parameter-informed evaluation: Our metrics are parameterized based on distributions observed in human-annotated itineraries, enabling more realistic and nuanced quality assessments.

Through extensive evaluation, we demonstrate that parameter-informed prompting strategies can substantially improve temporal scheduling quality, with 7-day trip meal scores improving from 0.61 to 0.80. However, we also reveal important trade-offs between itinerary quality and constraint satisfaction, highlighting the challenges in balancing optimization objectives in real-world travel planning scenarios.

*TripCraft*establishes a new standard for evaluating AI-powered travel planning systems, providing researchers and practitioners with a realistic, comprehensive benchmark that captures the complexity and constraints of real-world travel itinerary generation.

2 Related Work

2.1 Travel Planning Benchmarks

Several benchmarks have been developed to evaluate automated travel planning systems. TravelPlanner (Xie et al., 2024) introduced a comprehensive framework for real-world planning with language agents, focusing on multi-constraint satisfaction across flights, accommodations, and attractions. However, TravelPlanner relies heavily on semi-synthetic data and employs primarily binary constraint-checking metrics that may not capture nuanced quality differences between itineraries.

TravelAgent (Chen et al., 2024) explored personalized travel planning with AI assistants, emphasizing user preference modeling and recommendation quality. While this work highlighted the importance of personalization, it lacked comprehensive evaluation metrics for spatial and temporal coherence.

ChinaTravel (Shao et al., 2024) provided a region-specific benchmark for Chinese travel planning, incorporating cultural and geographical constraints specific to that context. However, like other existing benchmarks, it focuses on constraint satisfaction rather than continuous quality assessment.

These existing benchmarks, while valuable, share common limitations: (1) reliance on semi-synthetic or incomplete real-world data, (2) binary evaluation metrics that miss nuanced quality differences, and (3) insufficient integration of real-world constraints such as public transit schedules and event timing.

2.2 LLM Planning Evaluation

The evaluation of LLM planning capabilities has received significant attention in recent work. Pallagani et al. (Pallagani et al., 2023) provided a comprehensive analysis of LLM planning capabilities, highlighting both strengths and limitations in structured reasoning tasks. However, most evaluation frameworks focus on abstract planning problems rather than real-world applications with complex constraints.

Recent work has emphasized the importance of continuous rather than binary metrics for evaluating AI systems in complex domains. This shift toward nuanced evaluation reflects the recognition that binary pass/fail metrics often fail to capture meaningful quality differences that matter in practical applications.

2.3 Spatio-Temporal Reasoning

Travel planning inherently requires sophisticated spatio-temporal reasoning, considering both geographical constraints and temporal dependencies. Existing work in this area has typically focused on either spatial optimization (route planning) or temporal scheduling separately, rather than jointly optimizing both dimensions while considering user preferences and real-world constraints.

Our work bridges these gaps by providing a benchmark that integrates real-world data sources, employs continuous evaluation metrics, and jointly evaluates spatial, temporal, and personalization aspects of travel planning quality.

3 *TripCraft*

3.1 Benchmark Overview

*TripCraft*is designed as a comprehensive benchmark for evaluating spatio-temporally fine-grained travel planning capabilities. The benchmark consists of 1,000 carefully constructed travel queries spanning 140 U.S. cities, with varying trip durations (3, 5, and 7 days) and difficulty levels based on data availability and constraint complexity.

Figure 1 provides an overview of the *TripCraft*benchmark architecture, illustrating the integration of diverse data sources, constraint types, and evaluation methodologies. Our benchmark is built around the principle that effective travel planning requires simultaneous optimization

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across multiple dimensions: temporal scheduling, spatial efficiency, constraint satisfaction, and personalization alignment.

3.2 Data Sources and Collection

Table 1 shows the scale of our data collection effort, which integrates multiple real-world data sources to ensure comprehensive coverage and authenticity:

Database	Data Entries (#)
City Set	140
Flights	3,446,829
Restaurants	3,892
Attractions	5,043
Accommodations	2,800
Events	21,980
Nearest Transit Stop	8,723
Distance Matrix	19,460

Table 1: Data entries in the database have been scraped to include the most recent data.

Points of Interest (PoIs): We collected 5,043 attractions and 3,892 restaurants through web scraping and OpenStreetMap (OSM) integration, ensuring comprehensive coverage across our target cities. Each attraction is categorized into one of 15 predefined types (Table 2 shows a subset), with associated duration estimates derived from expert knowledge and crowd-sourced data.

Transportation Data: A critical innovation in *TripCraft*is the integration of GTFS (General Transit Feed Specification) public transit data covering all 140 cities in our dataset. This includes 8,723 transit stop locations and comprehensive route/schedule information, enabling realistic evaluation of transit-dependent itineraries.

Events and Activities: We incorporated 21,980 events across four categories (Sports, Arts & Theatre, Music, and Film), providing temporal anchors that add complexity and realism to travel planning scenarios.

Category	Duration (hrs)
Boat Tours & Water Sports	3.5
Casinos & Gambling	2.5
Museums	3.0
Nature & Parks	4.5
Nightlife	2.5
Sights & Landmarks	3.0

Table 2: Attraction visiting duration (hrs) for a subset of categories (due to page limit). The complete list of categories has been given in Table 8 of Appendix.

Accommodation and Flights: Following TravelPlanner's approach, we adapted flight data with proper date alignment and collected accommodation information across all target cities, ensuring complete trip planning coverage.

3.3 Query Construction and Personas

Each query in *TripCraft*includes rich persona information designed to capture the diversity of real-world travelers. Table 3 outlines the key components of our constraint and persona framework:

Queries are systematically generated with randomized departure and destination cities, travel dates, and constraint combinations. City coverage scales with trip duration: 1-2 cities for 3-day trips, 2-3 cities for 5-day trips, and up to 3 cities for 7-day trips. This ensures appropriate complexity scaling while maintaining realistic travel patterns.

The dataset distribution across difficulty levels is shown in Table 4, with difficulty determined by data availability, constraint density, and geographical complexity:

3.4 Annotation Protocol

We employed 25 graduate-level annotators with travel planning experience to create gold standard itineraries. The annotation process involved multiple stages:

Initial Planning: Annotators created detailed day-by-day itineraries considering all provided constraints and persona requirements, with an average completion time of 30 minutes per instance.

Rationale Documentation: Each itinerary includes detailed rationales explaining scheduling decisions, POI selections, and persona alignment choices.

Expert Review: A subset of annotations underwent expert review combining automated validation scripts and manual quality checks to ensure consistency and quality.

Iterative Refinement: Based on expert feedback, annotation guidelines were refined through multiple iterations to improve consistency and coverage.

Figure 2 illustrates the complete data construction pipeline, from initial web scraping through final verification:

3.5 Evaluation Metrics

*TripCraft*introduces five novel continuous evaluation metrics that provide nuanced assessment of

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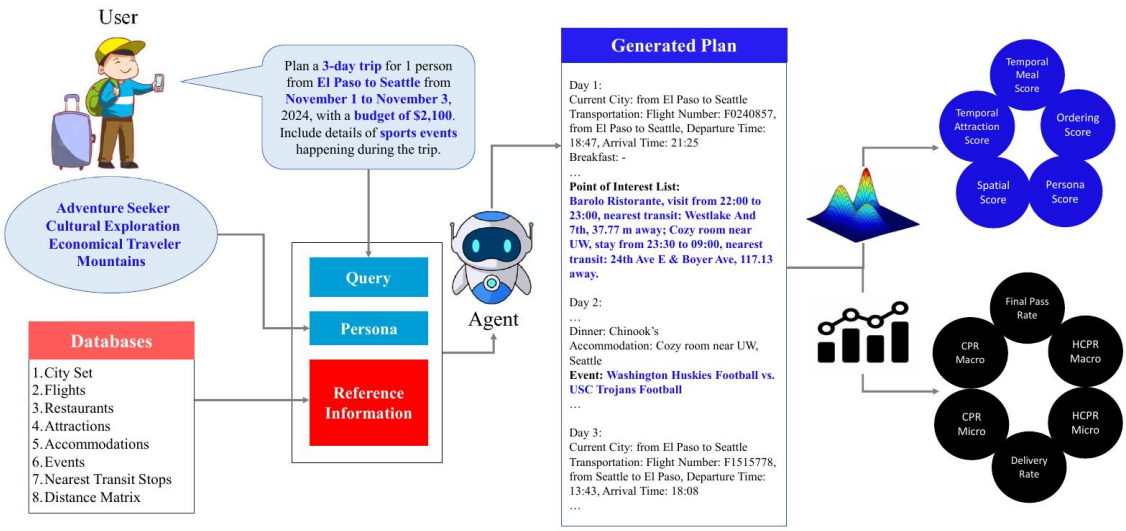

Figure 1: *TripCraft*overview: Continuous evaluation metrics along with fine-grained itinerary details.

Commonsense Constraints	
Sufficient Meal Gaps	Defines a minimum gap of four hours between the meals.
Valid PoI list	Defines validity rules for the Point of Interest list.
Diverse Events	Event choices should not be repeated throughout the trip.
Hard Constraints	
Event Types	Event Types include four distinct categories—Sports, Arts & Theatre, Music, and Film.
Attraction Types	Each attraction belongs to one or more of 15 predefined categories, ensuring a well-distributed selection of activities.
Persona Components	
Traveler Type	Defines how a traveler approaches their journey—whether they seek relaxation in cozy spots or adrenaline-pumping adventures.
Purpose of Travel	Captures trip motivation. Examples: to unwind, explore cultures etc.
Spending Preference	Reflects the traveler's budget and style, from luxurious indulgence to cost-conscious experiences.
Location Preference	Highlights preferred environments, such as beaches, mountains, cities, or wildlife-rich forests.

Table 3: Addition of constraints and persona details based on availability and limitations of scraped data has been given in the above table. The full list of constraints used in *TripCraft* has been given in Table 9 of Appendix.

Days	Easy	Medium	Hard	Total
3-day	130	106	108	344
5-day	87	164	73	324
7-day	99	178	55	332

Table 4: *TripCraft* dataset distribution

itinerary quality beyond binary constraint satisfaction:

Temporal Meal Score ($\bar{T}_{meal}$): Evaluates the appropriateness of meal timing based on natural dining patterns, using distributions derived from human annotations. This metric considers both absolute timing and spacing between meals.

Temporal Attraction Score ($\bar{T}_{attrac}$): Assesses whether attraction visit durations align with user pace preferences (e.g., "laid-back" vs. "adventurous") and activity types, using pace-specific duration distributions.

Spatial Score ($\bar{S}_{spatial}$): Measures the accessibility of planned activities via public transit, incorporating actual GTFS route data and walking distances to transit stops.

Ordering Score ($\bar{S}_{ord}$): Evaluates the efficiency of activity sequencing, considering geographical

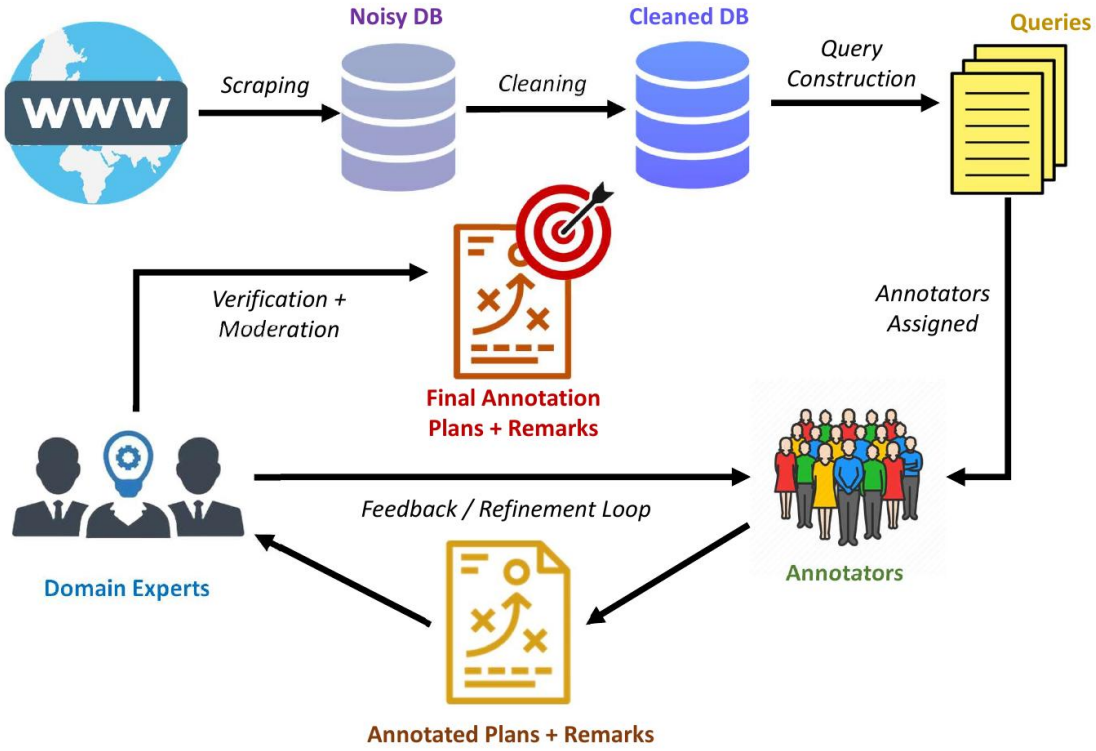

Figure 2: Visualization of the Construction Pipeline.

proximity and logical flow of activities throughout each day.

Persona Score ($\bar{S}_{persona}$): Quantifies alignment between selected activities and stated user preferences, including traveler type, purpose, spending level, and location preferences.

3.6 Parameter Calibration

All metrics are parameterized based on distributions observed in our human-annotated gold standard itineraries. Table 5 shows example parameters derived from annotation statistics:

These parameters ensure that our evaluation metrics reflect realistic expectations and preferences as demonstrated by human travel planners, rather than arbitrary thresholds.

4 Experiments and Results

4.1 Experimental Setup

We evaluate state-of-the-art language models on *TripCraft*using two primary settings: (1) standard prompting without parameter information, and (2) parameter-informed prompting that includes explicit guidance about meal timing, attraction durations, and other empirically-derived parameters from our annotation process.

Parameter	3-day	5-day	7-day
Restaurant Parameters			
Breakfast			
Mean Duration (hrs)	0.90	1.08	0.85
Std. Duration (hrs)	0.24	1.43	0.23
Lunch			
Mean Duration (hrs)	1.11	1.10	0.99
Std. Duration (hrs)	0.36	0.35	0.26
Dinner			
Mean Duration (hrs)	1.19	1.32	1.15
Std. Duration (hrs)	0.43	0.91	1.15
Attraction Parameters			
$\lambda_{laidback}$	1.10	1.26	1.11
$\lambda_{adventurous}$	2.01	1.61	1.82
σ_d (hrs)	1.11	1.07	0.90
n^{max}	5	4	4
n^{min}	0	0	0
k (hrs)	0.28	0.28	0.28

Table 5: Day-wise parameter details calculated from our annotation distribution. For a complete list of parameters refer to the Appendix.

The parameter-informed approach incorporates distributional information about meal timing preferences, attraction duration expectations for differ-

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ent traveler types, and spatial proximity requirements derived from our human annotation analysis. This allows us to investigate whether providing explicit parameter guidance improves itinerary quality while maintaining constraint satisfaction.

4.2 Continuous Metric Results

Table 6 presents results for our novel continuous evaluation metrics across both experimental settings:

Temporal Meal Score: Parameter-informed prompting yields substantial improvements across all trip durations. The most dramatic improvement occurs for 7-day trips, where scores increase from 0.61 to 0.80 (+31%). Shorter trips also show significant gains: 3-day trips improve from 0.56 to 0.70 (+25%) and 5-day trips from 0.69 to 0.75 (+9%). This demonstrates that explicit temporal guidance helps models generate more realistic meal timing.

Temporal Attraction Score: Results show mixed patterns across trip durations. While 3-day and 5-day trips show modest improvements (0.0169→0.0171 and 0.0057→0.0078), 7-day trips actually decrease (0.0122→0.0057). This suggests that parameter guidance may create tension between different optimization objectives in longer trips.

Spatial Score: Consistent improvements are observed across all trip durations, with gains ranging from 2-4%. This indicates that parameter-informed prompting helps models generate more spatially coherent itineraries with better transit accessibility.

Persona Score: Modest but consistent improvements across all trip durations (2-4% gains), suggesting that parameter information helps models better align itineraries with user preferences, though the effect size is smaller than for temporal metrics.

Ordering Score: Substantial improvements for shorter trips (3-day: 0.70→0.74, 5-day: 0.91→0.93) with minimal change for 7-day trips (0.96→0.97), indicating that parameter guidance particularly benefits sequential planning in less complex scenarios.

4.3 Traditional Constraint Results

Table 7 shows results for traditional binary constraint satisfaction metrics:

Delivery Rate: While 3-day trips show improved delivery rates with parameter information (92.60%→96.08%), both 5-day and 7-day trips show decreased performance (98.69%→91.30% and 96.03%→92.51% respectively). This suggests a trade-off between itinerary quality optimization and basic feasibility.

Commonsense Pass Rate (CPR): Mixed results across trip durations. Micro-averaged CPR improves for 3-day trips (77.61%→80.08%) but decreases for longer trips (5-day: 66.99%→61.69%, 7-day: 70.35%→66.43%). Macro-averaged CPR shows the opposite pattern, suggesting different constraint types are affected differently.

Hard Constraint Pass Rate (HCPR): Generally improved performance with parameter information, particularly for 5-day and 7-day trips in micro-averaged metrics, though 3-day trips show slight decreases.

Final Pass Rate: Substantial decreases for 3-day and 5-day trips with parameter information, while 7-day trips show no change (both achieve 0% final pass rate). This highlights the challenge of satisfying all constraints simultaneously while optimizing for quality.

4.4 Quality vs. Feasibility Trade-offs

Our results reveal a fundamental trade-off between itinerary quality (as measured by our continuous metrics) and constraint satisfaction (as measured by traditional binary metrics). Parameter-informed prompting consistently improves temporal, spatial, and persona alignment but often at the cost of reduced constraint satisfaction rates.

This trade-off is particularly pronounced in longer trips, where the complexity of maintaining constraint satisfaction while optimizing multiple quality dimensions becomes increasingly challenging. The results suggest that current LLMs struggle to simultaneously optimize for all objectives, highlighting an important limitation in existing planning approaches.

Figure 3 illustrates this trade-off with concrete examples, showing how our continuous metrics capture quality differences that traditional binary metrics miss:

4.5 Analysis and Discussion

The experimental results demonstrate several key insights:

Parameter guidance effectiveness: Providing explicit parameter information derived from human annotations substantially improves temporal scheduling quality, with particularly strong effects on meal timing appropriateness.

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Setting	Category	$\bar{T}_{meal}$	$\bar{T}_{attrac}$	$\bar{S}_{spatial}$	$\bar{S}_{persona}$	$\bar{S}_{ord}$
w/o Parameter Info	3-day	0.56	0.0169	0.80	0.46	0.70
	5-day	0.69	0.0057	0.86	0.49	0.91
	7-day	0.61	**0.0122**	0.85	0.50	0.96
w/ Parameter Info	3-day	**0.70**	**0.0171**	**0.83**	**0.48**	**0.74**
	5-day	**0.75**	**0.0078**	**0.87**	**0.50**	**0.93**
	7-day	**0.80**	0.0057	**0.86**	**0.51**	**0.97**

Table 6: Results of our proposed evaluation metrics for the two settings. For each category, the **bolded** values represent the better ones between the two settings.

Setting	Category	Delivery Rate	CPR		HCPR		Final Pass Rate
			Micro	Macro	Micro	Macro	
w/o Parameter Info	3-day	92.60	77.61	**6.08**	**27.63**	**26.08**	**3.47**
	5-day	**98.69**	**66.99**	**2.17**	4.21	3.04	**1.74**
	7-day	**96.03**	**70.35**	0.00	0.62	0.00	0.00
w/ Parameter Info	3-day	**96.08**	**80.08**	5.21	26.10	25.21	1.74
	5-day	91.30	61.69	0.87	**8.60**	**5.21**	0.43
	7-day	92.51	66.43	0.00	**0.83**	0.00	0.00

Table 7: Results of existing evaluation metrics for the two settings. CPR and HCPR stand for Commonsense Pass Rate and Hard Constraint Pass Rate respectively. For each category, the **bolded** values represent the better ones between the two settings. Moreover, equal values have been underlined.

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Metric interdependencies: Different quality metrics show varying responsiveness to parameter guidance, suggesting that some aspects of itinerary planning are more amenable to explicit optimization than others.

Complexity scaling challenges: The trade-off between quality and feasibility becomes more pronounced in longer trips, indicating that current approaches may not scale well to complex multi-day planning scenarios.

Evaluation methodology importance: Our continuous metrics reveal quality differences that traditional binary metrics fail to capture, demonstrating the value of nuanced evaluation approaches for complex planning tasks.

5 Conclusion

We have introduced *TripCraft*, a comprehensive benchmark for evaluating spatio-temporally fine-grained travel planning that addresses critical limitations in existing evaluation frameworks. Our benchmark makes several key contributions to the field of AI-powered travel planning:

Real-world data integration: By incorporating GTFS public transit data, comprehensive event calendars, and detailed attraction categorizations across 140 U.S. cities, *TripCraft*provides a realistic foundation for evaluating travel planning systems. This integration of authentic real-world constraints represents a significant advancement over previous semi-synthetic benchmarks.

Continuous evaluation metrics: Our five novel metrics—Temporal Meal, Temporal Attraction, Spatial, Ordering, and Persona scores—provide nuanced quality assessment that captures important differences missed by traditional binary constraint-checking approaches. These metrics are parameterized based on distributions from expert human annotations, ensuring realistic evaluation standards.

Empirical insights: Through extensive evaluation, we have demonstrated that parameter-informed prompting can substantially improve specific aspects of itinerary quality, particularly temporal scheduling, with 7-day trip meal scores improving from 0.61 to 0.80. However, our results also reveal important trade-offs between quality optimization and constraint satisfaction, highlighting challenges in current LLM planning approaches.

Benchmark scale and diversity: With 1,000 queries spanning multiple trip durations, difficulty levels, and persona types, *TripCraft*provides comprehensive coverage for evaluating travel planning systems across diverse scenarios.

Our experimental results reveal that while current language models can generate reasonable travel itineraries, there remains significant room for improvement in simultaneously optimizing mul-

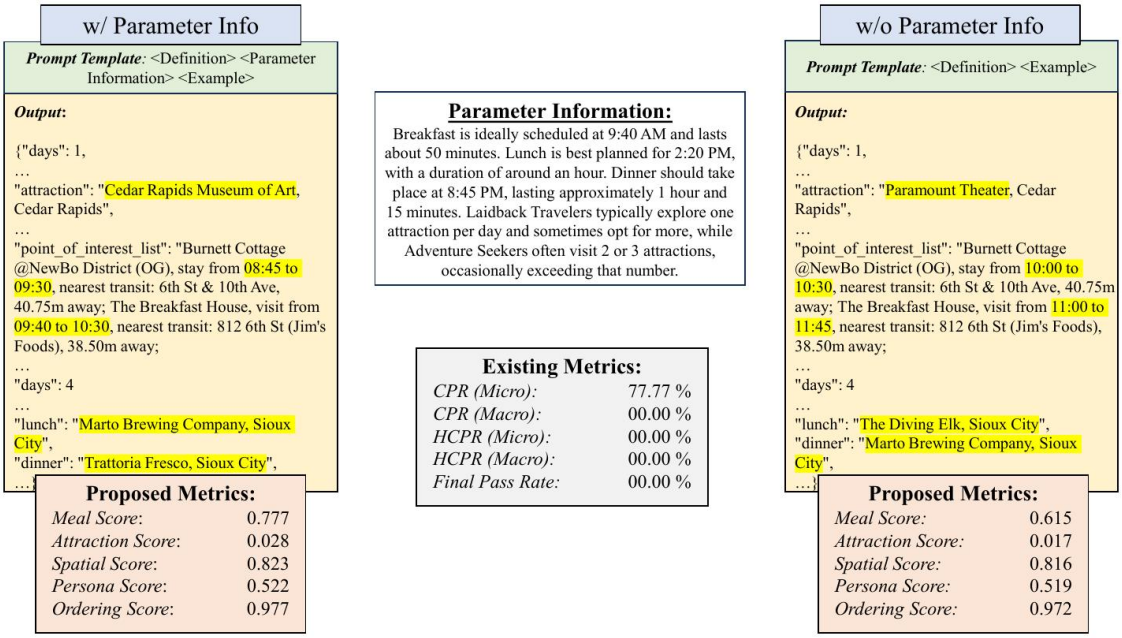

Figure 3: Existing constraint-based metrics rate these plans equally, but our metrics provide a continuous score, capturing temporal, spatial, sequential, and personal differences.

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tiple quality dimensions while maintaining constraint satisfaction. The trade-offs we observe between itinerary quality and feasibility highlight important research directions for future work.

5.1 Limitations and Future Work

While *TripCraft*represents a significant advancement in travel planning evaluation, several limitations suggest directions for future research:

Geographic coverage: Our current focus on U.S. cities limits the cultural and geographical diversity of planning scenarios. Future work should expand to international destinations with different cultural norms and transportation systems.

Temporal scope: The benchmark currently focuses on 3-7 day trips. Extending to longer journeys or different temporal patterns (e.g., weekend getaways, extended travel) would broaden applicability.

Dynamic constraints: Real-world travel planning must often adapt to changing conditions (weather, closures, etc.). Incorporating dynamic constraint scenarios would increase realism.

Multi-traveler scenarios: Current personas focus on individual travelers. Group travel planning presents additional complexity that warrants investigation.

Evaluation methodology: While our continuous metrics represent an advancement, further research into human judgment correlation and metric validation would strengthen the evaluation framework.

*TripCraft*establishes a new standard for realistic, comprehensive evaluation of AI travel planning systems. We hope this benchmark will facilitate research progress toward more capable, practical travel planning agents that can handle the complexity and constraints of real-world travel scenarios. The benchmark and evaluation tools will be made publicly available to support future research in this important application domain.

The integration of real-world constraints, continuous quality metrics, and expert annotations in *TripCraft*provides a foundation for advancing the state-of-the-art in automated travel planning while highlighting important challenges that must be addressed to achieve practical deployment of such systems.

References

Aili Chen, Xuyang Ge, Ziquan Fu, Yanghua Xiao, and Jiangjie Chen. 2024. Travelagent: An ai assistant for personalized travel planning. Preprint, arXiv:2409.08069.

Vishal Pallagani, Bharath Muppasani, Keerthiram Murugesan, Francesca Rossi, Biplav Srivastava, Lior Horesh, Francesco Fabiano, and Andrea Loreggia. 2023. Understanding the capabilities of large lan-

guage models for automated planning. arXiv preprint arXiv:2305.16151.

Jie-Jing Shao, Xiao-Wen Yang, Bo-Wen Zhang, Baizhi Chen, Wen-Da Wei, Lan-Zhe Guo, and Yu-feng Li. 2024. Chinatravel: A real-world benchmark for language agents in chinese travel planning. arXiv preprint arXiv:2412.13682.

Jian Xie, Kai Zhang, Jiangjie Chen, Tinghui Zhu, Renze Lou, Yuandong Tian, Yanghua Xiao, and Yu Su. 2024. Travelplanner: A benchmark for real-world planning with language agents. In Forty-first International Conference on Machine Learning.

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